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1. Register Complaint (Manual Testing)
2. Run Black Box Tests
3. Run White Box Tests
4. Exit
Select an option: 2
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Test Case: TC1
Description: Verify successful complaint registration

Input:
  Customer ID: C101
  Complaint: Internet not working
  System Available: true
  Agent Available: true

Expected:
  Complaint recorded successfully
  SLA assigned
  Assigned to agent
  Confirmation sent

Actual:
  Complaint recorded successfully
  SLA assigned
  Assigned to agent
  Confirmation sent

Match: PASS
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Test Case: TC2
Description: Reject invalid customer

Input:
  Customer ID: C999
  Complaint: Speed issue
  System Available: true
  Agent Available: true

Expected:
  Customer validation failed

Actual:
  Customer validation failed: Unknown customer ID.
  No complaint record created

Match: PASS
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Test Case: TC3

Description: No complaint record for invalid customer

Input:

Customer ID: C999
Complaint: Billing issue
System Available: true
Agent Available: true

Expected:

Customer validation failed
No complaint record created

Actual:

Customer validation failed: Unknown customer ID.
No complaint record created

Match: PASS

Test Case: TC4

Description: Store complaint successfully

Input:

Customer ID: C102
Complaint: Router reboot required
System Available: true
Agent Available: true

Expected:

Complaint recorded successfully

Actual:

Complaint recorded successfully
SLA assigned
Assigned to agent
Confirmation sent

Match: PASS

Test Case: TC5

Description: Assign SLA for valid complaint

Input:

Customer ID: C103
Complaint: Frequent disconnections
System Available: true
Agent Available: true

Expected:
SLA assigned

Actual:
Complaint recorded successfully
SLA assigned
Assigned to agent
Confirmation sent

Match: PASS

Test Case: TC6
Description: Assign agent when available

Input:
Customer ID: C101
Complaint: No signal
System Available: true
Agent Available: true

Expected:
Assigned to agent

Actual:
Complaint recorded successfully
SLA assigned
Assigned to agent
Confirmation sent

Match: PASS

Test Case: TC7
Description: Send confirmation for valid flow

Input:
Customer ID: C102
Complaint: Slow speed
System Available: true
Agent Available: true

Expected:
Confirmation sent

Actual:
Complaint recorded successfully
SLA assigned
Assigned to agent
Confirmation sent

Match: PASS

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Test Case: TC8  
Description: Reject missing complaint details  
  
Input:  
  Customer ID: C101  
  Complaint:  
  System Available: true  
  Agent Available: true  
  
Expected:  
  Complaint validation failed  
  
Actual:  
  Complaint validation failed: Complaint details are required.  
  
Match: PASS  
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Test Case: TC9  
Description: Show retry message when system is unavailable  
  
Input:  
  Customer ID: C101  
  Complaint: Line down  
  System Available: false  
  Agent Available: true  
  
Expected:  
  System unavailable  
  
Actual:  
  System unavailable. Please try again later.  
  
Match: PASS  
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Test Case: TC10  
Description: Escalate when no agent is available  
  
Input:  
  Customer ID: C102  
  Complaint: Installation pending  
  System Available: true  
  Agent Available: false  
  
Expected:  
  Escalated to supervisor  
  
Actual:  
  Complaint recorded successfully  
  SLA assigned
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Escalated to supervisor
Confirmation sent

Match: PASS
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Test Case: TC11
Description: SLA assigned before agent allocation

Input:
  Customer ID: C103
  Complaint: Packet loss
  System Available: true
  Agent Available: true

Expected:
  Complaint recorded successfully
  SLA assigned
  Assigned to agent

Actual:
  Complaint recorded successfully
  SLA assigned
  Assigned to agent
  Confirmation sent

Match: PASS
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Test Case: TC12
Description: Stop flow after validation failure

Input:
  Customer ID: C999
  Complaint: Device issue
  System Available: true
  Agent Available: true

Expected:
  Customer validation failed
  No complaint record created

Actual:
  Customer validation failed: Unknown customer ID.
  No complaint record created

Match: PASS
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Test Case: TC13
Description: Assign unique identifiers

Input:
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Input:

Customer ID: C101
Complaint: Service outage
System Available: true
Agent Available: true

Expected:

Complaint IDs are unique

Actual:

Complaint IDs are unique

Match: PASS

Test Case: TC14

Description: Process complaints independently

Input:

Customer ID: C101
Complaint: Payment update
System Available: true
Agent Available: true

Expected:

Complaints processed independently

Actual:

Complaints processed independently

Match: PASS

Test Case: TC15

Description: Complete flow executed successfully

Input:

Customer ID: C102
Complaint: New connection request
System Available: true
Agent Available: true

Expected:

Complaint recorded successfully
SLA assigned
Assigned to agent
Confirmation sent

Actual:

Complaint recorded successfully
SLA assigned
Assigned to agent
Confirmation sent

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Actual:
  Complaint recorded successfully
  SLA assigned
  Assigned to agent
  Confirmation sent

Match: PASS
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Total Test Cases: 15
Passed: 15
Failed: 0
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